

# IMST sentire™ Radar Module

## 24 GHz sR-1200 Series

### Interface Description

Version 1.7



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## About this Document

This document contains a user command description of the IMST's sentire™ Radar Module. The following table displays the changes applied to this manual and assigns a version number to each document release.

| Version | Date of Change | Reason of Change                                                                    |
|---------|----------------|-------------------------------------------------------------------------------------|
| 1.0     | 2016-04-20     | initial document                                                                    |
| 1.1     | 2016-06-22     | Range Finder algorithm added                                                        |
| 1.2     | 2016-07-07     | Description of self-test commands modified                                          |
| 1.3     | 2017-01-19     | Added description of: Automatic Operation Mode, Firmware Update, Software Extension |
| 1.4     | 2017-12-07     | Added description of HTTP server                                                    |
| 1.5     | 2018-03-22     | Added description of HTTP server                                                    |
| 1.6     | 2018-05-24     | Added command to unlock full frequency band                                         |
| 1.7     | 2018-05-28     | Description of HT parameters modified                                               |
|         |                |                                                                                     |



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# 1. Introduction

This manual describes the communication of the IMST's sentire<sup>TM</sup> Radar Module (further called radar module or simply radar) with the control system (host). The radar operates in FMCW or CW mode at a frequency of 24 GHz and provides the capability for range and velocity measurements in a wide area of applications. Using the integrated transmitter and receiver antennas, it is also possible to estimate the angle-of-arrival of the back-scattered signal, corresponding to a certain target direction. The radar module is available with three different interfaces (SPI, CAN or Ethernet). In case of the SPI interface, an external SPI-to-USB adapter allows to connect the radar directly to the computer. More information regarding the interface hardware or the radar connection protocol is available from the Radar User Manual /RAD/.

Independently from the selected interface, the module provides an extensive set of user commands for the following tasks:

- configuration of radar parameters, as the operation mode or the frequency range,
- accessing internal radar parameters, as the current frontend temperature or the transmission power,
- triggering a measurement process and accessing the generated data,
- starting high-level applications, e.g. the Human Tracker algorithm.

Additionally, the GUI program IMST SenTool is optionally provided as part of the Development Kit (see /GUI/). This software is based on the command protocol described in this document and offers the possibility to modify the most of the radar parameters, to make different measurements as well as to visualize and save measurement data. The IMST SenTool is very helpful to explore the possibilities of the sentire<sup>TM</sup> Radar Module or to determine suitable settings for a particular application or operation.



## 2. Description of the Communication Protocol

The communication is initiated by the host. The host sends a command, which is performed by the radar module. To every command belongs a certain dialog (data transferred from and to the host). When the command is done, the radar module returns to waiting state.

### 2.1 Communication Interfaces

The communication interface provides the connection between the radar-module and the host. Currently three possible interfaces are used:

- SPI (see section 2.1.1)
- CAN Bus (see section 2.1.2)
- Ethernet (see section 2.1.3)

#### 2.1.1 SPI

When connected via SPI interface the radar-module operates in slave mode. The host must be configured as master. Table 2.1 shows the recommended settings. Please note that one host can be connected to several radar modules using different COM-ports.

| No. | Parameter                       | Value           |
|-----|---------------------------------|-----------------|
| 1   | SPI Clock Rate                  | 0.1 – 2 MHz     |
| 2   | SPI Chip Select                 | low active      |
| 3   | SPI Master Transmit Data        | on falling edge |
| 4   | SPI Master Receive Data         | on rising edge  |
| 5   | SPI Rx/Tx Word length           | 16 Bits         |
| 6   | SPI Byte Order (read and write) | LSB first       |
| 7   | SPI Mode                        | Master          |

Table 2.1: SPI configuration parameters

### 2.1.2 CAN Bus

When connected via CAN bus interface the radar can be addressed using its node ID (which should be unique in the CAN network) as well as a broadcast ID which is identical for all radar modules. After a successful connection, the CAN bus parameters of a radar module can be accessed and optionally modified as described in sections 2.5.14 and 2.5.15. Table 2.2 summarizes the CAN bus configuration parameters.

One host can be connected to several radar modules, when they have different node IDs or baud rates. Communication over the broadcast node goes to all modules.

| No. | Parameter                   | Value                                     |
|-----|-----------------------------|-------------------------------------------|
| 1   | Baud Rate [kbaud]           | 10, 20, 50, 100, 125, 250, 500, 800, 1000 |
| 2   | Default Baud Rate           | 1000                                      |
| 3   | Node ID                     | 0 to 62 (default 2)                       |
| 4   | Broadcast node ID           | 63                                        |
| 5   | Rx/Tx Word length           | 16 Bits                                   |
| 6   | Byte Order (read and write) | LSB first                                 |

Table 2.2: CAN bus configuration parameters

### 2.1.3 Ethernet

When connected via Ethernet interface the radar can be addressed using its Ethernet configuration parameters (see table 2.3). After a successful connection, the communication parameters of a radar module can be accessed and optionally modified as described in sections 2.5.14 and 2.5.15. One host can be connected to several radar modules, when they have different connection parameters (port and IP address).

The default connection parameters are valid 10 seconds after power-up. After this time the radar module is reconnected with the user-defined connection parameters.

If the user-defined connection parameters are not known, they can be requested by con-

necting during the start phase with the default parameters and sending an interface parameter request.

| No. | Parameter          | Value       |
|-----|--------------------|-------------|
| 1   | Default Port       | 1024        |
| 2   | Default IP-Address | 192.168.0.2 |
| 3   | Gateway            | 192.168.0.1 |
| 4   | Subnet Mask        | 255.255.0.0 |

Table 2.3: Ethernet configuration parameters

## 2.2 Data Format

The logical data can be single values of 1, 2, 4, 8-byte length. The physical data transmission depends on the used interface.

**Note:** Some interfaces require a minimum size of 2 bytes per value. When the protocol describes a single-byte value, this value has to be padded, when one of these interfaces is used.

### 2.2.1 SPI

The SPI is configured to transmit word-length (2 bytes) messages. The low byte is transmitted first. If the data protocol requires transmission of a single byte, 0x00 is sent as high byte.

### 2.2.2 CAN Bus

The CAN bus is configured to transmit frames with extended identifiers (header size is 29 bit, see table 2.4). Since every frame holds up to 8 byte data, the payload has to be split into 8-byte segments. If necessary the last segment has to be padded with zeros.

The data length is multiple of 2 bytes, the least significant bytes first. Single-byte data are padded with 0x00 as high byte.

| Bit   | Field Name               | Description                                                               |
|-------|--------------------------|---------------------------------------------------------------------------|
| 0–7   | Command ID               | Lower byte of the transmitted command ID                                  |
| 8–10  | Reserved                 | don't care                                                                |
| 11–16 | CAN node ID              | Node ID of the addressed radar module                                     |
| 17    | Protocol/Application Bit | 0: application message, 1: protocol message (currently not in use)        |
| 18–21 | Sequence Number          | Sequence number of the currently transmitted sub-message, starting with 0 |
| 22    | Final Bit                | 1: last sub-message, 0: else                                              |
| 23    | First Bit                | 1: first sub-message, 0: else                                             |
| 24    | Direction Bit            | 0: sensor to host, 1: host to sensor                                      |
| 25–27 | Message Type             | 0: SYNC, 1: REQ, 2: RSP, 3: IND, 4: ACK (currently not in use)            |
| 28    | Reserved                 | must be always 1                                                          |

Table 2.4: Header definition of a CAN message

### 2.2.3 Ethernet

The Ethernet interface can transmit single bytes. The least significant bytes of a multi-byte value are transmitted first.

## 2.3 Command Execution Protocol

The start and the end of a message are identified by an interface-dependent handshake scheme. Each command is identified by a 16-bit integer value, called *Command ID*. The further data flow and the actions performed by the radar depend on this command ID.

After processing the received command and executing the corresponding instructions, the module indicates the end of communication by another interface-dependent handshake.

### 2.3.1 SPI

In SPI mode, the pattern 0xAA55 is used for synchronization purposes as well as for indication of data transmission. The next word received by the radar module is the command ID (2 bytes). After the execution of command-dependent instructions, the synchronization word and the command ID are sent back to the host as acknowledgment. Table 2.5 summarizes the data flow in SPI interface mode.

| Step | Transmission from Host to Radar   | Transmission from Radar to Host                                         |
|------|-----------------------------------|-------------------------------------------------------------------------|
| 1    | Synchronization pattern 0xAA55    |                                                                         |
| 2    | Command ID (2 bytes)              |                                                                         |
| 3    | Command dependent data (optional) |                                                                         |
| 4    |                                   | Synchronization pattern 0xAA55<br>(only if followed by additional data) |
| 5    |                                   | Command dependent data (optional)                                       |
| 6    |                                   | Synchronization pattern 0xAA55                                          |
| 7    |                                   | Command ID (2 bytes)                                                    |

Table 2.5: Data flow in SPI mode

### 2.3.2 CAN Bus

The underlying CAN bus protocol provides handshake and command identification, so no additional synchronization pattern is required. Every segment of a CAN message contains the command ID, and the first and the last segments are specially marked (the complete definition of a CAN message header is given by table 2.4).

**Note:** After executing the command-dependent instructions, the radar will *always* transmit a CAN message with the corresponding command ID, even if no data are expected by the host. This final message is indicated by a *Final Bit* of 1 and acts as concluding handshake.

### 2.3.3 Ethernet

As the communication of the Ethernet interface is based on TCP, no additional synchronization pattern is required. The first received word is interpreted by the radar as command ID. As acknowledgment, the command ID is immediately sent back to the host before the execution of command-dependent instructions. After the command, no additional communication takes place. Table 2.6 summarizes the data flow in Ethernet interface mode.

| Step | Transmission from Host to Radar   | Transmission from Radar to Host   |
|------|-----------------------------------|-----------------------------------|
| 1    | Command ID (2 bytes)              |                                   |
| 2    |                                   | Command ID (2 bytes)              |
| 3    | Command dependent data (optional) |                                   |
| 4    |                                   | Command dependent data (optional) |

Table 2.6: Data flow in Ethernet mode

## 2.4 Command Overview

Table 2.7 shows an overview of the commands explained in this document. A detailed description of all user commands can be found in chapters 2.5 to 2.7.

| Command ID | Description                                    | Chapter | Page |
|------------|------------------------------------------------|---------|------|
| 0x0000     | Ping (communication test)                      | 2.5.1   | 10   |
| 0x0001     | Get system time                                | 2.5.2   | 10   |
| 0x0002     | Set system time                                | 2.5.3   | 11   |
| 0x0012     | Send system information                        | 2.5.4   | 11   |
| 0x0013     | Self test                                      | 2.5.5   | 13   |
| 0x0014     | Initialize self test                           | 2.5.6   | 14   |
| 0x0020     | Set operation parameters                       | 2.5.7   | 14   |
| 0x0021     | Get operation parameters                       | 2.5.8   | 15   |
| 0x0027     | Reset system parameters                        | 2.5.9   | 16   |
| 0x0028     | Set system parameters                          | 2.5.10  | 16   |
| 0x0029     | Get system parameters                          | 2.5.11  | 18   |
| 0x002B     | Get radiated power and sensor temperature      | 2.5.12  | 19   |
| 0x002C     | Restart CPU                                    | 2.5.13  | 20   |
| 0x0030     | Get user-defined interface parameters          | 2.5.14  | 20   |
| 0x0031     | Set user-defined interface parameters          | 2.5.15  | 21   |
| 0x0032     | Activate user-defined interface parameters     | 2.5.16  | 22   |
| 0x0040     | FMCW measurement with UP-ramp                  | 2.5.17  | 23   |
| 0x0041     | FMCW measurement with DOWN-ramp                | 2.5.18  | 23   |
| 0x0042     | CW measurement                                 | 2.5.19  | 24   |
| 0x0043     | Get time-domain data                           | 2.5.20  | 24   |
| 0x0044     | Get frequency-domain data                      | 2.5.21  | 25   |
| 0x0045     | Get time-domain data after UP-ramp             | 2.5.22  | 27   |
| 0x0046     | Get frequency-domain data after UP-ramp        | 2.5.23  | 27   |
| 0x0047     | Get time-domain data after DOWN-ramp           | 2.5.24  | 28   |
| 0x0048     | Get frequency-domain data after DOWN-ramp      | 2.5.25  | 28   |
| 0x0049     | Get time-domain data after CW-measurement      | 2.5.26  | 29   |
| 0x004A     | Get frequency-domain data after CW-measurement | 2.5.27  | 30   |
| 0x0060     | Set Human Tracker parameters                   | 2.6.1   | 31   |
| 0x0062     | Get Human Tracker detection threshold          | 2.6.4   | 33   |
| 0x0063     | Do Human Tracker measurement                   | 2.6.3   | 32   |
| 0x0065     | Get Human Tracker parameters                   | 2.6.2   | 32   |

| Command ID | Description                  | Chapter | Page |
|------------|------------------------------|---------|------|
| 0x0066     | Set Range Finder parameters  | 2.7.1   | 35   |
| 0x0067     | Get Range Finder parameters  | 2.7.2   | 36   |
| 0x0068     | Do Range Finder measurement  | 2.7.4   | 37   |
| 0x006A     | Reset Range Finder algorithm | 2.7.3   | 36   |
| 0x00FF     | Software Extension           | 2.10    | 45   |

Table 2.7: Overview of radar user commands

## 2.5 Basic Commands

The following chapters are describing the dialog between host and radar module according to the command ID.

The data is described by abstract data types (e.g. Uint16: unsigned integer 16-bit). The byte order for each interface is described in chapter 2.2. It has to be noted that one-byte data elements (Uint8) have to be padded, if the interface is configured for 2-byte data.

### 2.5.1 Communication Test

This command checks the communication between radar-module and host. There is no payload.

|                                            |                               |             |
|--------------------------------------------|-------------------------------|-------------|
| Command Name                               | CMDID_PING                    |             |
| Command ID                                 | 0x0000                        |             |
| Description                                | Performs a communication test |             |
| <i>Data transmitted from Host to Radar</i> |                               |             |
| Name                                       | Data Type                     | Description |
| None                                       |                               |             |
| <i>Data transmitted from Radar to Host</i> |                               |             |
| Name                                       | Data Type                     | Description |
| None                                       |                               |             |

Table 2.8: Description of the command CMDID\_PING

### 2.5.2 Get System Time

This command reads the radar system time counter in milliseconds.

After power-up the counter is 0 (representing 1.1.1970 0:00). The CPU actualizes the counter by an internal timer. Thus, when not synchronized, the counter merely represents the time after power up.

When synchronized with the host's system time the counter represents the time since the UNIX epoch. Most programming languages provide standard library functions for conversion from and to human-readable time/date-format.

|                                            |                                |                                                      |
|--------------------------------------------|--------------------------------|------------------------------------------------------|
| Command Name                               | CMDID_GETTIME                  |                                                      |
| Command ID                                 | 0x0001                         |                                                      |
| Description                                | Send radar system time to host |                                                      |
| <i>Data transmitted from Host to Radar</i> |                                |                                                      |
| Name                                       | Data Type                      | Description                                          |
| None                                       |                                |                                                      |
| <i>Data transmitted from Radar to Host</i> |                                |                                                      |
| Name                                       | Data Type                      | Description                                          |
| Milliseconds                               | Uint64                         | Current system time in milliseconds since UNIX epoch |

Table 2.9: Description of the command CMDID\_GETTIME

### 2.5.3 Set System Time

This command synchronizes the radar system time with the host clock.

|                                     |                                          |                                                       |
|-------------------------------------|------------------------------------------|-------------------------------------------------------|
| Command Name                        | CMDID_SETTIME                            |                                                       |
| Command ID                          | 0x0002                                   |                                                       |
| Description                         | Set radar system time to a defined value |                                                       |
| Data transmitted from Host to Radar |                                          |                                                       |
| Name                                | Data Type                                | Description                                           |
| Milliseconds                        | Uint64                                   | Current system time in milliseconds since UNIX epoch. |
| Data transmitted from Radar to Host |                                          |                                                       |
| Name                                | Data Type                                | Description                                           |
| None                                |                                          |                                                       |

Table 2.10: Description of the command CMDID\_SETTIME

### 2.5.4 Send System Information

This command is used to read the system information of the radar module.

|                                            |                                                |                                                                                                                                                                                                                                                                    |
|--------------------------------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Command Name                               | CMDID_SEND_INFO                                |                                                                                                                                                                                                                                                                    |
| Command ID                                 | 0x0012                                         |                                                                                                                                                                                                                                                                    |
| Description                                | Send hardware and software information to host |                                                                                                                                                                                                                                                                    |
| <i>Data transmitted from Host to Radar</i> |                                                |                                                                                                                                                                                                                                                                    |
| Name                                       | Data Type                                      | Description                                                                                                                                                                                                                                                        |
| None                                       |                                                |                                                                                                                                                                                                                                                                    |
| <i>Data transmitted from Radar to Host</i> |                                                |                                                                                                                                                                                                                                                                    |
| Name                                       | Data Type                                      | Description                                                                                                                                                                                                                                                        |
| Firmware Version                           | Uint32                                         | Firmware version (e.g. 0x00040102 corresponds to 4.1.2)                                                                                                                                                                                                            |
| Firmware Revision                          | Uint32                                         | Firmware revision: special customer related firmware releases are additionally identified by a revision number (e.g. 92957)                                                                                                                                        |
| SNT Board Version                          | Uint32                                         | Identification of the used hardware (versions of the built-in PCBs, only for internal use)                                                                                                                                                                         |
| Baseband Board Version                     | Uint32                                         |                                                                                                                                                                                                                                                                    |
| Frontend Board Version                     | Uint32                                         |                                                                                                                                                                                                                                                                    |
| Available Channels                         | Uint16                                         | Bits 0 – 15 represent available receiver channels Rx1 to Rx16                                                                                                                                                                                                      |
| Available Algorithms                       | Uint16                                         | Bits 0 – 15 represent available algorithms <ul style="list-style-type: none"><li>• Bit 0 and 1: Raw data algorithms</li><li>• Bit 2: obsolete</li><li>• Bit 3: obsolete</li><li>• Bit 4: Human Tracker algorithm</li><li>• Bit 5: Range Finder algorithm</li></ul> |
| Radar Hardware ID                          | Uint16                                         | Components used in the radar hardware (for internal use)                                                                                                                                                                                                           |
| Radar Number                               | Uint32                                         | Unique radar identifier                                                                                                                                                                                                                                            |
| Flashing Date                              | Uint32                                         | Date of flashing (e.g. 20150301 stands for March, 1st 2015)                                                                                                                                                                                                        |
| Phase Offset                               | Int32                                          | Phase correction value [deg] for view angle estimation                                                                                                                                                                                                             |

Table 2.11: Description of the command CMDID\_SEND\_INFO

### 2.5.5 Perform Self Test

This command initiates a built-in self test of the radar module. The result is an integer number representing an error code.

**Note:** When several errors occur, the error codes are OR-combined.

|                                            |                                                        |                                           |
|--------------------------------------------|--------------------------------------------------------|-------------------------------------------|
| Command Name                               | CMDID_SELF_TEST                                        |                                           |
| Command ID                                 | 0x0013                                                 |                                           |
| Description                                | Performs a series of self-tests and returns the result |                                           |
| <i>Data transmitted from Host to Radar</i> |                                                        |                                           |
| Name                                       | Data Type                                              | Description                               |
| None                                       |                                                        |                                           |
| <i>Data transmitted from Radar to Host</i> |                                                        |                                           |
| Name                                       | Data Type                                              | Description                               |
| Error Code                                 | Uint32                                                 | Corresponding error code (see table 2.13) |

Table 2.12: Description of the command CMDID\_SELF\_TEST

| Error ID     | Error Code | Description                                                                   |
|--------------|------------|-------------------------------------------------------------------------------|
| EC_NO        | 0x00000000 | No error detected                                                             |
| EC_CPU       | 0x00000001 | Test Central processing Unit (CPU) of the radar module                        |
| EC_RAM       | 0x00000002 | Test Random Access Memory (RAM) of the radar module                           |
| EC_FLASH     | 0x00000004 | Test Flash memory of the radar module                                         |
| EC_RAD_CHIP  | 0x00000008 | Test Tx/Rx radar chip of the radar module                                     |
| EC_VOLT_HIGH | 0x00000010 | Test voltage of the radar power supply                                        |
| EC_VOLT_LOW  | 0x00000020 |                                                                               |
| EC_CURR_HIGH | 0x00000040 | Test current consumed by the radar module (currently not active)              |
| EC_CURR_LOW  | 0x00000080 |                                                                               |
| EC_TEMP_HIGH | 0x00000100 | Test internal temperature of the radar module                                 |
| EC_TEMP_LOW  | 0x00000200 |                                                                               |
| EC_RF_HIGH   | 0x00000400 | Test transmitted RF power of the radar module                                 |
| EC_RF_LOW    | 0x00000800 |                                                                               |
| EC_DDS       | 0x00001000 | Test signal generation by the Direct Digital Synthesis (currently not active) |
| EC_PLL       | 0x00002000 | Test state of the Phase Locked Loop (PLL)                                     |

| Error ID | Error Code | Description                                                                    |
|----------|------------|--------------------------------------------------------------------------------|
| EC_DAC   | 0x00004000 | Test D/A converter of the radar module (currently not active)                  |
| EC_I2C   | 0x00008000 | Test I <sup>2</sup> C communication of the radar module (currently not active) |
| EC_CH1_I | 0x00010000 | Test all receiver channels of the radar module                                 |
| EC_CH1_Q | 0x00020000 |                                                                                |
| EC_CH2_I | 0x00040000 |                                                                                |
| EC_CH2_Q | 0x00080000 |                                                                                |

Table 2.13: List of self-test error codes

### 2.5.6 Initialize Self Test

This command initializes the self-test of the radar module by calculating a checksum over the Flash memory and writing this value to EEPROM. By executing the Flash memory test (corresponding to error code EC\_FLASH), the radar module recalculates this checksum and compares it with the reference value.

**Note:** The command CMDID\_INIT\_SELF\_TEST shall be executed just one time after each modification of the radar's firmware. Therefore, the parameter *AdvancedEn* must be set to 1 (refer to section 2.5.10).

|                                     |                                          |             |
|-------------------------------------|------------------------------------------|-------------|
| Command Name                        | CMDID_INIT_SELF_TEST                     |             |
| Command ID                          | 0x0014                                   |             |
| Description                         | Initialize self-test of the radar module |             |
| Data transmitted from Host to Radar |                                          |             |
| Name                                | Data Type                                | Description |
| None                                |                                          |             |
| Data transmitted from Radar to Host |                                          |             |
| Name                                | Data Type                                | Description |
| None                                |                                          |             |

Table 2.14: Description of the command CMDID\_INIT\_SELF\_TEST

### 2.5.7 Set Operation Parameters

With this command the operation parameters can be set. These parameters are used in the Automatic Operation mode described in section 2.8 which is only available for the Ethernet interface.

|                                                 |                          |             |
|-------------------------------------------------|--------------------------|-------------|
| Command Name                                    | CMDID_OP_SETUP           |             |
| Command ID                                      | 0x0020                   |             |
| Description                                     | Set operation parameters |             |
| <i>Data transmitted from Host to Radar</i>      |                          |             |
| Name                                            | Data Type                | Description |
| Operation parameters (as defined in table 2.16) |                          |             |
| <i>Data transmitted from Radar to Host</i>      |                          |             |
| Name                                            | Data Type                | Description |
| None                                            |                          |             |

Table 2.15: Description of the command CMDID\_OP\_SETUP

| <b>Name</b>  | <b>Data Type</b> | <b>Description</b>                                                                                        |
|--------------|------------------|-----------------------------------------------------------------------------------------------------------|
| opMode       | Uint16           | Operation mode: 0 for normal operation triggered by the host, 1 to enable sockets for automatic operation |
| autoMeasType | Uint16           | Bitmask that determines the type of measurement. Refer to section 2.8                                     |
| autoDelay    | Uint16           | Delay in [ms] between two cycles of measurement and data transmission                                     |

Table 2.16: Description of the operation parameters

### 2.5.8 Get Operation Parameters

With this command the current set operation parameters can be requested. These parameters are used in the Automatic Operation mode described in section 2.8 which is only available for the Ethernet interface.

|                                                 |                          |             |
|-------------------------------------------------|--------------------------|-------------|
| Command Name                                    | CMDID_SEND_OP            |             |
| Command ID                                      | 0x0021                   |             |
| Description                                     | Get operation parameters |             |
| <i>Data transmitted from Host to Radar</i>      |                          |             |
| Name                                            | Data Type                | Description |
| None                                            |                          |             |
| <i>Data transmitted from Radar to Host</i>      |                          |             |
| Name                                            | Data Type                | Description |
| Operation parameters (as defined in table 2.16) |                          |             |

Table 2.17: Description of the command CMDID\_SEND\_OP

### 2.5.9 Reset System Parameters

This command causes the user defined system parameters to be deleted. The module will update the system with an internally defined default parameter set.

**Note:** This command should only be used in case of emergency.

|                                            |                                                                |             |
|--------------------------------------------|----------------------------------------------------------------|-------------|
| Command Name                               | CMDID_RST_PARAMS                                               |             |
| Command ID                                 | 0x0027                                                         |             |
| Description                                | Reset system parameters of the radar module to a default state |             |
| <i>Data transmitted from Host to Radar</i> |                                                                |             |
| Name                                       | Data Type                                                      | Description |
| None                                       |                                                                |             |
| <i>Data transmitted from Radar to Host</i> |                                                                |             |
| Name                                       | Data Type                                                      | Description |
| None                                       |                                                                |             |

Table 2.18: Description of the command CMDID\_RST\_PARAMS

### 2.5.10 Set System Parameters

This command is used to set the most important parameters for radar measurements. The values are saved in an EEPROM, so they are available after a power-up of the radar module. The system parameters may be read-only (R/O) or read-write (R/W). This command transmits only the R/W-parameters.

**Note:** If valid values for the parameters *MinFreq* and *ManualBW* are specified, the parameter *band* will not be taken into account. The adjustable frequencies are internally limited to 23–26 GHz. Please note that this frequency range is not guaranteed since the real possible range depends on the fabrication process of the used hardware. To use one of the predefined frequency bands these parameters can be set to e.g. 0.

|                                                    |                                          |             |
|----------------------------------------------------|------------------------------------------|-------------|
| Command Name                                       | CMDID_SETUP                              |             |
| Command ID                                         | 0x0028                                   |             |
| Description                                        | Restart radar with new system parameters |             |
| <i>Data transmitted from Host to Radar</i>         |                                          |             |
| Name                                               | Data Type                                | Description |
| Radar system parameters (as defined in table 2.20) |                                          |             |
| <i>Data transmitted from Radar to Host</i>         |                                          |             |
| Name                                               | Data Type                                | Description |
| None                                               |                                          |             |

Table 2.19: Description of the command CMDID\_SETUP

| Name        | Data Type | Description                                                                                                                                                                                                                                   |
|-------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Band        | UInt8     | Bandwidth selection (see table 2.21)                                                                                                                                                                                                          |
| TRamp       | UInt8     | Ramp time in [ms]                                                                                                                                                                                                                             |
| ZeroPad     | UInt8     | Zero pad factor (1 = no padding, 2, 4, 8)                                                                                                                                                                                                     |
| FftDataType | UInt8     | Type of transmitted frequency-domain data <ul style="list-style-type: none"> <li>• 0: magnitude only</li> <li>• 1: magnitude and phase [rad]</li> <li>• 2: real and imaginary parts</li> <li>• 3: magnitude and object angle [rad]</li> </ul> |
| FrontendEn  | UInt8     | Flag: frontend power enabled (0 = OFF, 1 = ON)                                                                                                                                                                                                |
| PowerSaveEn | UInt8     | Flag: power saving mode enabled (0 = OFF, 1 = ON), if ON the frontend power is switched OFF between the pulses                                                                                                                                |
| Norm        | UInt8     | Flag: signal normalization enabled                                                                                                                                                                                                            |
| ActChannels | UInt16    | Bits 0 – 15 represent active receiver channels                                                                                                                                                                                                |
| AdvancedEn  | UInt8     | Flag: Advanced mode enabled (only for internal use)                                                                                                                                                                                           |

| Name        | Data Type | Description                                                                                                                                                                                                                                                                                                                                 |
|-------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FreqPoints  | Uint16    | Number frequency-domain values to be calculated and sent (number of FD samples), valid are values from 5 to 513                                                                                                                                                                                                                             |
| MinFreq     | Uint16    | Start frequency in [MHz], if out of limits, the Band parameters will be used                                                                                                                                                                                                                                                                |
| ManualBW    | Uint16    | Bandwidth in [MHz], if out of limits, the Band parameters will be used                                                                                                                                                                                                                                                                      |
| Attenuation | Uint8     | Number that represents an attenuation value in [dB] for the Tx output power: 0 $\approx$ 0 dB, 1 $\approx$ 0.4 dB, 2 $\approx$ 0.8 dB, 3 $\approx$ 1.4 dB, 4 $\approx$ 2.5 dB, 5 $\approx$ 4 dB, 6 $\approx$ 6 dB, 7 $\approx$ 9 dB. <b>Note</b> that the resulting adjusted attenuation is dependent on the temperature and the frequency. |

Table 2.20: Description of system parameters

| Band | Bandwidth [MHz] | Start-Frequency [GHz] | Stop-Frequency [GHz] |
|------|-----------------|-----------------------|----------------------|
| 0    | 50              | 24.000                | 24.050               |
| 1    | 75              | 24.000                | 24.075               |
| 2    | 125             | 24.000                | 24.125               |
| 3    | 230             | 24.000                | 24.230               |
| 4    | 250             | 24.000                | 24.250               |
| 5    | 500             | 24.000                | 24.500               |
| 6    | 800             | 24.000                | 24.800               |
| 7    | 1000            | 24.000                | 25.000               |
| 8    | 1200            | 24.000                | 25.200               |
| 9    | 1500            | 24.000                | 25.500               |
| 10   | 1800            | 24.000                | 25.800               |
| 11   | 2000            | 23.500                | 25.500               |
| 12   | 2200            | 23.500                | 25.700               |
| 13   | 2500            | 23.200                | 25.700               |

Table 2.21: Radar bandwidth selection

### 2.5.11 Get System Parameters

This command requests all status-information to the host. With this command the R/W-parameters are transmitted first and then the R/O-parameters.

Since the R/O-parameters are recalculated after each *Set System Parameters* command it is recommended to read back the system parameters after every set up. Additional information can be found in section A.1.

|                                                    |                                                               |                                 |
|----------------------------------------------------|---------------------------------------------------------------|---------------------------------|
| Command Name                                       | CMDID_SEND_PARAMS                                             |                                 |
| Command ID                                         | 0x0029                                                        |                                 |
| Description                                        | Get a list of current system parameters from the radar module |                                 |
| <i>Data transmitted from Host to Radar</i>         |                                                               |                                 |
| Name                                               | Data Type                                                     | Description                     |
| None                                               |                                                               |                                 |
| <i>Data transmitted from Radar to Host</i>         |                                                               |                                 |
| Name                                               | Data Type                                                     | Description                     |
| Radar system parameters (as defined in table 2.20) |                                                               |                                 |
| RangeBin                                           | UInt32                                                        | Size of a Range bin in [μm]     |
| DopplerBin                                         | UInt32                                                        | Size of a Doppler bin in [μm/s] |
| FrequencyBin                                       | UInt32                                                        | Size of a Frequency bin in [Hz] |

Table 2.22: Description of the command CMDID\_SEND\_PARAMS

### 2.5.12 Get Radiated Power and Sensor Temperature

This command returns the Tx output power measured in the used transceiver chip, the temperature of this chip and the temperature measured on the RF frontend. The value for the Tx output power is transmitted in [dBm/10].

|                                     |                                                      |                                             |
|-------------------------------------|------------------------------------------------------|---------------------------------------------|
| Command Name                        | CMDID_SEND_SENSORDATA                                |                                             |
| Command ID                          | 0x002B                                               |                                             |
| Description                         | Get internal sensor temperature and current Tx power |                                             |
| Data transmitted from Host to Radar |                                                      |                                             |
| Name                                | Data Type                                            | Description                                 |
| None                                |                                                      |                                             |
| Data transmitted from Radar to Host |                                                      |                                             |
| Name                                | Data Type                                            | Description                                 |
| NtcTemp                             | Int32                                                | Current sensor temperature in [°C]          |
| TxPower                             | UInt32                                               | Current transmitted power in [dBm/10]       |
| ChipTemp                            | Int32                                                | Temperature of the transceiver chip in [°C] |

Table 2.23: Description of the command CMDID\_SEND\_SENSORDATA

### 2.5.13 Restart CPU

This command initiates a CPU restart. The command is for further use, if a boot loader is implemented.

|                                            |                   |                 |
|--------------------------------------------|-------------------|-----------------|
| Command Name                               | CMDID_RESTART_CPU |                 |
| Command ID                                 | 0x002C            |                 |
| Description                                | Restart CPU       |                 |
| <i>Data transmitted from Host to Radar</i> |                   |                 |
| Name                                       | Data Type         | Description     |
| Dummy                                      | Uint16            | Dummy Reception |
| <i>Data transmitted from Radar to Host</i> |                   |                 |
| Name                                       | Data Type         | Description     |
| None                                       |                   |                 |

Table 2.24: Description of the command CMDID\_RESTART\_CPU

### 2.5.14 Read Interface Parameters

This command reads the user-defined interface parameters from EEPROM. As there are no configurable parameters for the SPI interface, this command is available only for Ethernet and CAN bus. The data format depends on the selected interface:

- the Ethernet connection is defined by IP address and port (see table 2.25),
- the CAN Bus connection is defined by the node identifier and the baud rate (see table 2.26).

|                                            |                           |             |
|--------------------------------------------|---------------------------|-------------|
| Command Name                               | CMDID_GET_IFC_PARAMS      |             |
| Command ID                                 | 0x0030                    |             |
| Description                                | Read interface parameters |             |
| <i>Data transmitted from Host to Radar</i> |                           |             |
| Name                                       | Data Type                 | Description |
| None                                       |                           |             |
| <i>Data transmitted from Radar to Host</i> |                           |             |
| Name                                       | Data Type                 | Description |
| IP                                         | 4 × Uint8                 | IP address  |
| Port                                       | Uint16                    | Port number |

Table 2.25: Description of the command CMDID\_GET\_IFC\_PARAMS (Ethernet)

|                                     |                           |                              |
|-------------------------------------|---------------------------|------------------------------|
| Command Name                        | CMDID_GET_IFC_PARAMS      |                              |
| Command ID                          | 0x0030                    |                              |
| Description                         | Read interface parameters |                              |
| Data transmitted from Host to Radar |                           |                              |
| Name                                | Data Type                 | Description                  |
| None                                |                           |                              |
| Data transmitted from Radar to Host |                           |                              |
| Name                                | Data Type                 | Description                  |
| Node                                | Uint16                    | Node identifier (0–62)       |
| Speed                               | Uint16                    | Transmission speed in [kbps] |

Table 2.26: Description of the command CMDID\_GET\_IFC\_PARAMS (CAN bus)

### 2.5.15 Write Interface Parameters

This command writes new user-defined interface parameters to EEPROM. As there are no configurable parameters for the SPI interface, this command is available only for Ethernet (see table 2.27) and CAN bus (see table 2.28). The data format depends on the selected interface.

|                                            |                            |             |
|--------------------------------------------|----------------------------|-------------|
| Command Name                               | CMDID_SET_IFC_PARAMS       |             |
| Command ID                                 | 0x0031                     |             |
| Description                                | Write interface parameters |             |
| <i>Data transmitted from Host to Radar</i> |                            |             |
| Name                                       | Data Type                  | Description |
| IP                                         | 4 × Uint8                  | IP address  |
| Port                                       | Uint16                     | Port number |
| <i>Data transmitted from Radar to Host</i> |                            |             |
| Name                                       | Data Type                  | Description |
| None                                       |                            |             |

Table 2.27: Description of the command CMDID\_SET\_IFC\_PARAMS (Ethernet)

|                                     |                            |                              |
|-------------------------------------|----------------------------|------------------------------|
| Command Name                        | CMDID_SET_IFC_PARAMS       |                              |
| Command ID                          | 0x0031                     |                              |
| Description                         | Write interface parameters |                              |
| Data transmitted from Host to Radar |                            |                              |
| Name                                | Data Type                  | Description                  |
| Node                                | Uint16                     | Node identifier (0–62)       |
| Speed                               | Uint16                     | Transmission speed in [kbps] |
| Data transmitted from Radar to Host |                            |                              |
| Name                                | Data Type                  | Description                  |
| None                                |                            |                              |

Table 2.28: Description of the command CMDID\_SET\_IFC\_PARAMS (CAN bus)

### 2.5.16 Activate Interface Parameters

The user-defined interface parameters are permanently stored in EEPROM. They are activated after reset of the radar. The commands 0x0030 and 0x0031 (read and write interface parameters) just read or write the EEPROM values. The current connection parameters remain unchanged.

This command performs a reconnection with the user-defined parameters without triggering a reset. There is no payload.

|                                     |                                                                     |             |
|-------------------------------------|---------------------------------------------------------------------|-------------|
| Command Name                        | CMDID_ACTIVATE_IFC_PARAMS                                           |             |
| Command ID                          | 0x0032                                                              |             |
| Description                         | Performs a reconnection with the user-defined interface parameters. |             |
| Data transmitted from Host to Radar |                                                                     |             |
| Name                                | Data Type                                                           | Description |
| None                                |                                                                     |             |
| Data transmitted from Radar to Host |                                                                     |             |
| Name                                | Data Type                                                           | Description |
| None                                |                                                                     |             |

Table 2.29: Description of the command CMDID\_ACTIVATE\_IFC\_PARAMS

### 2.5.17 FMCW Measurement with UP-Ramp

With this command an UP-Ramp is performed. The Radar-Module generates a frequency-ramp from the lower to the upper band-limit (UP-Ramp) of the selected bandwidth (see table 2.19 on how to change bandwidth). The corresponding measurement data is generated but not transmitted to host.

|                                            |                                        |             |
|--------------------------------------------|----------------------------------------|-------------|
| Command Name                               | CMDID_UP_RAMP                          |             |
| Command ID                                 | 0x0040                                 |             |
| Description                                | Generate an UP-Ramp and sample IF data |             |
| <i>Data transmitted from Host to Radar</i> |                                        |             |
| Name                                       | Data Type                              | Description |
| None                                       |                                        |             |
| <i>Data transmitted from Radar to Host</i> |                                        |             |
| Name                                       | Data Type                              | Description |
| None                                       |                                        |             |

Table 2.30: Description of the command CMDID\_UP\_RAMP

### 2.5.18 FMCW Measurement with DOWN-Ramp

Analog to 2.5.17, a DOWN-Ramp is performed with this command. The corresponding measurement data is generated but not transmitted to host.

|                                            |                                          |             |
|--------------------------------------------|------------------------------------------|-------------|
| Command Name                               | CMDID_DN_RAMP                            |             |
| Command ID                                 | 0x0041                                   |             |
| Description                                | Generate an DOWN-Ramp and sample IF data |             |
| <i>Data transmitted from Host to Radar</i> |                                          |             |
| Name                                       | Data Type                                | Description |
| None                                       |                                          |             |
| <i>Data transmitted from Radar to Host</i> |                                          |             |
| Name                                       | Data Type                                | Description |
| None                                       |                                          |             |

Table 2.31: Description of the command CMDID\_DN\_RAMP

### 2.5.19 Perform CW-Measurement

With this command a measurement without ramp is performed. The corresponding measurement data is generated but not transmitted to host.

|                                            |                                     |             |
|--------------------------------------------|-------------------------------------|-------------|
| Command Name                               | CMDID_GAP                           |             |
| Command ID                                 | 0x0042                              |             |
| Description                                | Sample IF data at a fixed frequency |             |
| <i>Data transmitted from Host to Radar</i> |                                     |             |
| Name                                       | Data Type                           | Description |
| None                                       |                                     |             |
| <i>Data transmitted from Radar to Host</i> |                                     |             |
| Name                                       | Data Type                           | Description |
| None                                       |                                     |             |

Table 2.32: Description of the command CMDID\_GAP

### 2.5.20 Get Time-Domain Data

The Radar-Module will send Time-Domain data from the last measurement *without* doing a new FMCW or CW measurement.

|                                                  |                                              |                                                |
|--------------------------------------------------|----------------------------------------------|------------------------------------------------|
| Command Name                                     | CMDID_TDATA                                  |                                                |
| Command ID                                       | 0x0043                                       |                                                |
| Description                                      | Get Time-Domain data for all active channels |                                                |
| <i>Data transmitted from Host to Radar</i>       |                                              |                                                |
| Name                                             | Data Type                                    | Description                                    |
| None                                             |                                              |                                                |
| <i>Data transmitted from Radar to Host</i>       |                                              |                                                |
| Name                                             | Data Type                                    | Description                                    |
| NumActChan                                       | Uint16                                       | Total number of active Rx channels             |
| for channel = 0; channel < NumActChan; channel++ |                                              |                                                |
| for sample = 0; sample < 1024; sample++          |                                              |                                                |
| Data                                             | Int32                                        | Data sample of the digitized IF signal         |
| end                                              |                                              |                                                |
| end                                              |                                              |                                                |
| PreTime                                          | Uint32                                       | Time stamp before data transmission in [10 μs] |
| PostTime                                         | Uint32                                       | Time stamp after data transmission in [10 μs]  |

Table 2.33: Description of the command CMDID\_TDATA

### 2.5.21 Get Frequency-Domain Data

The Radar-Module will send FFT-Data in Cartesian or FFT-Data in polar coordinates from the last measurement *without* doing a new FMCW or CW measurement. The Data type which will be transmitted depends on the parameter *FftDatTyp* (see table 2.19).

|                                     |                                                                      |             |
|-------------------------------------|----------------------------------------------------------------------|-------------|
| Command Name                        | CMDID_FDATA                                                          |             |
| Command ID                          | 0x0044                                                               |             |
| Description                         | Calculate and receive Frequency-Domain data from all active channels |             |
| Data transmitted from Host to Radar |                                                                      |             |
| Name                                | Data Type                                                            | Description |
| None                                |                                                                      |             |
| Data transmitted from Radar to Host |                                                                      |             |
| Name                                | Data Type                                                            | Description |

|                                                         |        |                                                                                                                                                                                                                                                        |
|---------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FftDatTyp                                               | Uint8  | Type of transmitted frequency-domain data <ul style="list-style-type: none"> <li>• 0: magnitudes only</li> <li>• 1: magnitude and phase [rad]</li> <li>• 2: real and imaginary parts</li> <li>• 3: magnitude and view angle [rad]</li> </ul>           |
| PreProcTime                                             | Uint32 | Time stamp before data processing in [10 $\mu$ s]                                                                                                                                                                                                      |
| PostProcTime                                            | Uint32 | Time stamp after data processing in [10 $\mu$ s]                                                                                                                                                                                                       |
| <b>for</b> channel = 0; channel < 4; channel ++         |        |                                                                                                                                                                                                                                                        |
| DataMin                                                 | Int32  | Minimum TD value of the corresponding Rx channel                                                                                                                                                                                                       |
| <b>end</b>                                              |        |                                                                                                                                                                                                                                                        |
| <b>for</b> channel = 0; channel < 4; channel ++         |        |                                                                                                                                                                                                                                                        |
| DataMax                                                 | Int32  | Maximum TD value of the corresponding Rx channel                                                                                                                                                                                                       |
| <b>end</b>                                              |        |                                                                                                                                                                                                                                                        |
| Overload                                                | Uint8  | Number of first channel where an overload occurred                                                                                                                                                                                                     |
| NumActChan                                              | Uint16 | Total number of active Rx channels                                                                                                                                                                                                                     |
| FdSamples                                               | Uint16 | Number of Frequency-Domain data samples                                                                                                                                                                                                                |
| <b>for</b> channel = 0; channel < NumActChan; channel++ |        |                                                                                                                                                                                                                                                        |
| <b>for</b> sample = 0; sample < FdSamples; sample++     |        |                                                                                                                                                                                                                                                        |
| Data1                                                   | Int32  | FftDatTyp = <ul style="list-style-type: none"> <li>• 0, 1, 3: signal magnitude</li> <li>• 2: real part of the signal</li> </ul>                                                                                                                        |
| Data2                                                   | Int32  | FftDatTyp = <ul style="list-style-type: none"> <li>• 0: value is skipped</li> <li>• 1: signal phase in [rad] (IQ25 representation)</li> <li>• 2: imaginary part of the signal</li> <li>• 3: radar view angle in [rad] (IQ25 representation)</li> </ul> |
| <b>end</b>                                              |        |                                                                                                                                                                                                                                                        |
| <b>end</b>                                              |        |                                                                                                                                                                                                                                                        |

|          |        |                                                    |
|----------|--------|----------------------------------------------------|
| PostTime | Uint32 | Time stamp after data transmission in [10 $\mu$ s] |
|----------|--------|----------------------------------------------------|

Table 2.34: Description of the command CMDID\_FDATA

### 2.5.22 Get TD-Data after UP-Ramp

With this command a FMCW UP-Ramp is performed, the time-domain signal is sampled and the arising data is transmitted to the host. Details for this command can be found in Appendix A.2.

|                                             |                                                                         |             |
|---------------------------------------------|-------------------------------------------------------------------------|-------------|
| Command Name                                | CMDID_UP_RMP_TD                                                         |             |
| Command ID                                  | 0x0045                                                                  |             |
| Description                                 | Generate an UP-Ramp and receive Time-Domain data of all active channels |             |
| <i>Data transmitted from Host to Radar</i>  |                                                                         |             |
| Name                                        | Data Type                                                               | Description |
| None                                        |                                                                         |             |
| <i>Data transmitted from Radar to Host</i>  |                                                                         |             |
| Name                                        | Data Type                                                               | Description |
| Time-Domain data (as defined in table 2.33) |                                                                         |             |

Table 2.35: Description of the command CMDID\_UP\_RMP\_TD

### 2.5.23 Get FD-Data after UP-Ramp

With this command a FMCW UP-Ramp is performed, and the FFT-data of the reflected time-domain-signal is transmitted to the host. Details for this command can found in Appendix A.3.

|                                                  |                                                                              |             |
|--------------------------------------------------|------------------------------------------------------------------------------|-------------|
| Command Name                                     | CMDID_UP_RMP_FD                                                              |             |
| Command ID                                       | 0x0046                                                                       |             |
| Description                                      | Generate an UP-Ramp and receive Frequency-Domain data of all active channels |             |
| <i>Data transmitted from Host to Radar</i>       |                                                                              |             |
| Name                                             | Data Type                                                                    | Description |
| None                                             |                                                                              |             |
| <i>Data transmitted from Radar to Host</i>       |                                                                              |             |
| Name                                             | Data Type                                                                    | Description |
| Frequency-Domain data (as defined in table 2.34) |                                                                              |             |

Table 2.36: Description of the command CMDID\_UP\_RMP\_FD

### 2.5.24 Get TD-Data after DOWN-Ramp

Same as CMDID\_UP\_RMP\_TD in section 2.5.22 but with DOWN-Ramp instead of UP-Ramp.

|                                             |                                                                          |             |
|---------------------------------------------|--------------------------------------------------------------------------|-------------|
| Command Name                                | CMDID_DN_RMP_TD                                                          |             |
| Command ID                                  | 0x0047                                                                   |             |
| Description                                 | Generate a DOWN-Ramp and receive Time-Domain data of all active channels |             |
| <i>Data transmitted from Host to Radar</i>  |                                                                          |             |
| Name                                        | Data Type                                                                | Description |
| None                                        |                                                                          |             |
| <i>Data transmitted from Radar to Host</i>  |                                                                          |             |
| Name                                        | Data Type                                                                | Description |
| Time-Domain data (as defined in table 2.33) |                                                                          |             |

Table 2.37: Description of the command CMDID\_DN\_RMP\_TD

### 2.5.25 Get FD-Data after DOWN-Ramp

Same as CMDID\_UP\_RMP\_FD in section 2.5.23 but with DOWN-ramp instead of UP-ramp.

|                                                  |                                                                               |             |
|--------------------------------------------------|-------------------------------------------------------------------------------|-------------|
| Command Name                                     | CMDID_DN_RMP_FD                                                               |             |
| Command ID                                       | 0x0048                                                                        |             |
| Description                                      | Generate a DOWN-Ramp and receive Frequency-Domain data of all active channels |             |
| <i>Data transmitted from Host to Radar</i>       |                                                                               |             |
| Name                                             | Data Type                                                                     | Description |
| None                                             |                                                                               |             |
| <i>Data transmitted from Radar to Host</i>       |                                                                               |             |
| Name                                             | Data Type                                                                     | Description |
| Frequency-Domain data (as defined in table 2.34) |                                                                               |             |

Table 2.38: Description of the command CMDID\_DN\_RMP\_FD

### 2.5.26 Get TD-Data after CW-Measurement

The Radar-Module is also capable to perform CW-measurements. In a CW-measurement no frequency-ramp is driven when the time-domain signal is sampled. So, just a single frequency is emitted and only moving objects can be detected from the sampled data. This is based on the Doppler-effect.

This command initiates a CW-measurement and afterwards the Radar Module transmits the Time-Domain data back to the host.

|                                             |                                                                              |             |
|---------------------------------------------|------------------------------------------------------------------------------|-------------|
| Command Name                                | CMDID_GP_TD                                                                  |             |
| Command ID                                  | 0x0049                                                                       |             |
| Description                                 | Perform a CW measurement and receive Time-Domain data of all active channels |             |
| <i>Data transmitted from Host to Radar</i>  |                                                                              |             |
| Name                                        | Data Type                                                                    | Description |
| None                                        |                                                                              |             |
| <i>Data transmitted from Radar to Host</i>  |                                                                              |             |
| Name                                        | Data Type                                                                    | Description |
| Time-Domain data (as defined in table 2.33) |                                                                              |             |

Table 2.39: Description of the command CMDID\_GP\_TD

### 2.5.27 Get FD-Data after CW-Measurement

This command initiates a CW-measurement before transmitting FFT-Data, generated from the Time-Domain data, back to the host.

|                                                  |                                                                                   |             |
|--------------------------------------------------|-----------------------------------------------------------------------------------|-------------|
| Command Name                                     | CMDID_GP_FD                                                                       |             |
| Command ID                                       | 0x004A                                                                            |             |
| Description                                      | Perform a CW measurement and receive Frequency-Domain data of all active channels |             |
| <i>Data transmitted from Host to Radar</i>       |                                                                                   |             |
| Name                                             | Data Type                                                                         | Description |
| None                                             |                                                                                   |             |
| <i>Data transmitted from Radar to Host</i>       |                                                                                   |             |
| Name                                             | Data Type                                                                         | Description |
| Frequency-Domain data (as defined in table 2.34) |                                                                                   |             |

Table 2.40: Description of the command CMDID\_GP\_FD

## 2.6 Commands of the Human Tracker Algorithm

The Human Tracker Algorithm can be used to track moving objects. The algorithm suppresses static background objects, even if the background changes over time due to an adaptive background modeling. Therefore different parameters can be adjusted.

**Note:** The HT algorithm is developed for applications in which the radar operates at a fixed position, e.g. mounted to the wall of a building for perimeter surveillance. When mounted to a moving platform, the radar will show poor performance, as the background suppression cannot deal with highly dynamic scenes.

### 2.6.1 Set Human Tracker Parameters

This command sets up the Human Tracker Algorithm **and** starts a reference measurement of the background. This background measurement is used to build an initial model of the static background.

|                                                     |                                                                                                   |             |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------|
| Command Name                                        | CMDID_HT_PARAMS                                                                                   |             |
| Command ID                                          | 0x0060                                                                                            |             |
| Description                                         | Set parameters of the Human Tracker (HT) algorithm and perform a series of reference measurements |             |
| <i>Data transmitted from Host to Radar</i>          |                                                                                                   |             |
| Name                                                | Data Type                                                                                         | Description |
| Human Tracker parameters (as defined in table 2.42) |                                                                                                   |             |
| <i>Data transmitted from Radar to Host</i>          |                                                                                                   |             |
| Name                                                | Data Type                                                                                         | Description |
| None                                                |                                                                                                   |             |

Table 2.41: Description of the command CMDID\_HT\_PARAMS

| Name         | Data Type | Description                                                                                                                               |
|--------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------|
| NRefPulses   | Uint16    | Number of reference measurements before starting HT (e.g. 100)                                                                            |
| TimeInterval | Uint16    | Time interval between calibration pulses in [ms] (this parameter only has an effect during the reference measurement, otherwise obsolete) |
| NTargets     | Uint16    | Maximum number of simultaneously detected or tracked targets (e.g. 5)                                                                     |
| BackgrFilter | Uint16    | Time constant for background suppression (e.g. 32)                                                                                        |

| Name          | Data Type | Description                                                                                                        |
|---------------|-----------|--------------------------------------------------------------------------------------------------------------------|
| ThreshFilter  | Uint16    | Time constant of the adaptive detection threshold (e.g. 100)                                                       |
| OverThreshold | Uint16    | Factor between signal amplitude and detection threshold required for target detection (e.g. 30 corresponds to 3.0) |
| MinDistance   | Uint16    | Minimum target distance [m]                                                                                        |
| MinSeparation | Uint16    | Minimum separation between two consecutive targets [bins]                                                          |
| EnableTracker | Uint16    | 0: detect targets without tracking; 1: enable tracking capability                                                  |
| MaxRangeShift | Uint16    | Maximum target displacement between 2 pulses in [m]                                                                |
| MaxLostDetect | Uint16    | Maximum number of detection losses before a target is removed from the target list (e.g. 3)                        |

Table 2.42: Description of the Human Tracker parameters

### 2.6.2 Get Human Tracker Parameters

On this command the radar sends the current setup of the Human Tracker algorithm to the host.

|                                                     |                                                                      |             |
|-----------------------------------------------------|----------------------------------------------------------------------|-------------|
| Command Name                                        | CMDID_SEND_HT_PARAMS                                                 |             |
| Command ID                                          | 0x0065                                                               |             |
| Description                                         | Get a list of current Human Tracker parameters from the Radar Module |             |
| Data transmitted from Host to Radar                 |                                                                      |             |
| Name                                                | Data Type                                                            | Description |
| None                                                |                                                                      |             |
| Data transmitted from Radar to Host                 |                                                                      |             |
| Name                                                | Data Type                                                            | Description |
| Human Tracker parameters (as defined in table 2.42) |                                                                      |             |

Table 2.43: Description of the command CMDID\_SEND\_HT\_PARAMS

### 2.6.3 Do Human Tracker Measurement

This command first starts a FMCW measurement with an Up-Ramp. After the measurement the generated data is processed by the Human Tracker algorithm and sent to the

host.

|                                                |                                                                                        |                                                                    |
|------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Command Name                                   | CMDID_DO_HT                                                                            |                                                                    |
| Command ID                                     | 0x0063                                                                                 |                                                                    |
| Description                                    | Perform a Human Tracker measurement, process the data and send the results to the host |                                                                    |
| <i>Data transmitted from Host to Radar</i>     |                                                                                        |                                                                    |
| Name                                           | Data Type                                                                              | Description                                                        |
| None                                           |                                                                                        |                                                                    |
| <i>Data transmitted from Radar to Host</i>     |                                                                                        |                                                                    |
| Name                                           | Data Type                                                                              | Description                                                        |
| PreMeasTime                                    | Uint32                                                                                 | Time stamp before measurement in [10 μs]                           |
| PostMeasTime                                   | Uint32                                                                                 | Time stamp after measurement in [10 μs]                            |
| PreProcTime                                    | Uint32                                                                                 | Time stamp before data processing in [10 μs] (*)                   |
| PostProcTime                                   | Uint32                                                                                 | Time stamp after data processing in [10 μs]                        |
| NTargets                                       | Uint16                                                                                 | Number of detected targets                                         |
| for targets = 0; targets < NTargets; targets++ |                                                                                        |                                                                    |
| TrackID                                        | Uint16                                                                                 | Identifier of a particular target (1 – N)                          |
| Tracked                                        | Uint16                                                                                 | 0: new target detected; 1: target from list confirmed              |
| DetCnt                                         | Uint16                                                                                 | Lifetime of a target (1 for new targets, +1 at each confirmation)  |
| Level                                          | Uint32                                                                                 | Linear target level                                                |
| Distance                                       | Uint32                                                                                 | Distance of a particular target in [mm]                            |
| Angle                                          | Int32                                                                                  | Object angle of a particular target in [rad] (IQ25 representation) |
| end                                            |                                                                                        |                                                                    |

Table 2.44: Description of the command CMDID\_DO\_HT (\*the parameter *PreProcTime* is overwritten by 0 if the radar detects a significant change of the signal amplitude, e.g. caused by sensor vibration)

#### 2.6.4 Get Human Tracker Detection Threshold

An object is detected if it has a higher amplitude value than the adaptive detection threshold multiplied by the factor OverThreshold. This threshold contains an amplitude value for each possible Range bin. Range bins are separated in steps of the distance resolution which is dependent on the bandwidth. With this command the points of this detection threshold can be read out. The length of the transmitted data depends on

distance resolution and the adjusted maximum distance.

|                                            |                                                                                      |                                                                                                           |
|--------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Command Name                               | CMDID_SEND_REF_DATA                                                                  |                                                                                                           |
| Command ID                                 | 0x0062                                                                               |                                                                                                           |
| Description                                | Transmit current Human Tracker detection threshold from the Radar Module to the host |                                                                                                           |
| <i>Data transmitted from Host to Radar</i> |                                                                                      |                                                                                                           |
| Name                                       | Data Type                                                                            | Description                                                                                               |
| None                                       |                                                                                      |                                                                                                           |
| <i>Data transmitted from Radar to Host</i> |                                                                                      |                                                                                                           |
| Name                                       | Data Type                                                                            | Description                                                                                               |
| NCells                                     | Uint16                                                                               | Number of transmitted resolution cells (value should correspond to “FreqPoints” of the system parameters) |
| for cells = 0; cells < NCells; cells++     |                                                                                      |                                                                                                           |
| Thresh                                     | Int32                                                                                | Single sample of the detection threshold                                                                  |
| end                                        |                                                                                      |                                                                                                           |

Table 2.45: Description of the command CMDID\_SEND\_REF\_DATA

## 2.7 Commands of the Range Finder Algorithm

The Range Finder Algorithm can be used to measure the distance between the radar and a potentially moving target. In order to resolve the Range-Doppler-Ambiguity leading to inaccurate range measurements, a linear Kalman Filter with configurable parameters is applied. Additionally to the target's distance and velocity, the algorithms also measures the corresponding object angle.

### 2.7.1 Set Range Finder Parameters

This command sets up the Range Finder algorithm and resets the state of the internal Kalman filter.

|                                                    |                                                                                   |             |
|----------------------------------------------------|-----------------------------------------------------------------------------------|-------------|
| Command Name                                       | CMDID_RF_PARAMS                                                                   |             |
| Command ID                                         | 0x0066                                                                            |             |
| Description                                        | Set parameters of the Range Finder (RF) algorithm and reset internal filter state |             |
| <i>Data transmitted from Host to Radar</i>         |                                                                                   |             |
| Name                                               | Data Type                                                                         | Description |
| Range Finder parameters (as defined in table 2.47) |                                                                                   |             |
| <i>Data transmitted from Radar to Host</i>         |                                                                                   |             |
| Name                                               | Data Type                                                                         | Description |
| None                                               |                                                                                   |             |

Table 2.46: Description of the command CMDID\_RF\_PARAMS

| Name      | Data Type | Description                                                            |
|-----------|-----------|------------------------------------------------------------------------|
| RampDir   | Uint16    | Direction of the transmitted frequency ramp (1: UP-ramp, 2: DOWN-ramp) |
| MaxAccel  | Uint16    | Maximum target acceleration in [mm/s <sup>2</sup> ] (e.g. 5000)        |
| MinRange  | Uint16    | Minimum detection range in [mm] (e.g 0)                                |
| DetThresh | Int16     | Target detection threshold in [dBm <sup>2</sup> ] (e.g 0)              |
| MaxUncert | Uint16    | Maximum range measurement uncertainty in [mm] (e.g. 1000)              |
| TrackerEn | Uint16    | Enable or disable target tracking (1: enable, 0: disable)              |

Table 2.47: Description of the Range Finder parameters

### 2.7.2 Get Range Finder Parameters

On this command the radar sends the current setup of the Range Finder algorithm to the host.

|                                                    |                                                                     |             |
|----------------------------------------------------|---------------------------------------------------------------------|-------------|
| Command Name                                       | CMDID_SEND_RF_PARAMS                                                |             |
| Command ID                                         | 0x0067                                                              |             |
| Description                                        | Get a list of current Range Finder parameters from the Radar Module |             |
| <i>Data transmitted from Host to Radar</i>         |                                                                     |             |
| Name                                               | Data Type                                                           | Description |
| None                                               |                                                                     |             |
| <i>Data transmitted from Radar to Host</i>         |                                                                     |             |
| Name                                               | Data Type                                                           | Description |
| Range Finder parameters (as defined in table 2.47) |                                                                     |             |

Table 2.48: Description of the command CMDID\_SEND\_RF\_PARAMS

### 2.7.3 Reset Range Finder Algorithm

On this command the radar resets the internal state of the Range Finder algorithm.

|                                            |                                                        |             |
|--------------------------------------------|--------------------------------------------------------|-------------|
| Command Name                               | CMDID_RESET_RF                                         |             |
| Command ID                                 | 0x006A                                                 |             |
| Description                                | Reset the internal state of the Range Finder algorithm |             |
| <i>Data transmitted from Host to Radar</i> |                                                        |             |
| Name                                       | Data Type                                              | Description |
| None                                       |                                                        |             |
| <i>Data transmitted from Radar to Host</i> |                                                        |             |
| Name                                       | Data Type                                              | Description |
| None                                       |                                                        |             |

Table 2.49: Description of the command CMDID\_RESET\_RF

### 2.7.4 Do Range Finder Measurement

This command performs a single radar measurement and evaluates the measurement data corresponding to the specified Range Finder parameters (see table 2.47).

|                                            |                                                                                       |                                                                                       |
|--------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Command Name                               | CMDID_DO_RF                                                                           |                                                                                       |
| Command ID                                 | 0x0068                                                                                |                                                                                       |
| Description                                | Perform a Range Finder measurement, process the data and send the results to the host |                                                                                       |
| <i>Data transmitted from Host to Radar</i> |                                                                                       |                                                                                       |
| Name                                       | Data Type                                                                             | Description                                                                           |
| None                                       |                                                                                       |                                                                                       |
| <i>Data transmitted from Radar to Host</i> |                                                                                       |                                                                                       |
| Name                                       | Data Type                                                                             | Description                                                                           |
| Range                                      | Int32                                                                                 | Target distance in [mm] (0x7FFFFFFF: invalid measurement)                             |
| Angle                                      | Int32                                                                                 | Object angle of a target in [rad] (IQ25 representation)                               |
| Speed                                      | Int32                                                                                 | Target speed in [mm/s] (speed is not available if the parameter <i>TrackerEn</i> = 0) |
| Time                                       | UInt64                                                                                | System measurement time in [ms] (refer to sections 2.5.2 and 2.5.3)                   |

Table 2.50: Description of the command CMDID\_DO\_RF

## 2.8 Automatic Operation Mode

This section describes a special operation mode which is only available for the Ethernet interface. In this mode, the radar module operates automatically (without being triggered by user commands) and transmits measurement data to one or several connected TCP sockets. The operation mode performed by the radar module is defined by the operation parameters (see table 2.16).

### 2.8.1 Activation of Automatic Mode

The automatic operation mode is activated by the user command `CMDID_OP_SETUP` (refer to section 2.5.7) if the parameter *opMode* is set to 1 (see table 2.16). Setting the parameter *opMode* to 0 deactivates the automatic operation mode. As the operation parameters are written to EEPROM, the radar remembers this settings after powering down/up. At the first sight, there is no difference to the behavior in normal operation mode since the radar can still be connected to the user defined port and all regular commands work. The difference is, that the radar module opens six new TCP sockets with fixed port numbers and the same (user defined) IP address. These fixed port numbers are: 47475, 47476, 47477, 47478, 47479, 47480.

Once a TCP connection to one of these sockets is established, the radar module automatically starts operating and transmitting measurement data to this socket. If several connections are established, the same data is transmitted to *all* listeners. The activated operation mode is determined by the parameter *autoMeasType* (see table 2.16) which represents a 16-bit mask. The first two bits define the generated RF pulse, all other bits the type of requested measurement data (see table 2.51). It should be noted that any combination of these bits is possible, even if some combinations may produce nonsense results (e.g. CW mode plus Human Tracker).

| Bit No. | Decimal Value | Description                                            |
|---------|---------------|--------------------------------------------------------|
| 0       | 1             | Type of measurement: 0 = FMCW, 1 = CW                  |
| 1       | 2             | Ramp direction (FMCW only): 0 = UP-Ramp, 1 = DOWN-Ramp |
| 2       | 4             | System time in milliseconds (table 2.9)                |
| 3       | 8             | Temperatures and transmitted power (table 2.23)        |
| 4       | 16            | Header data (table 2.53)                               |
| 5       | 32            | Time domain data (table 2.33)                          |
| 6       | 64            | Frequency domain data (table 2.34)                     |
| 7       | 128           | Range Finder results (table 2.50)                      |
| 8       | 256           | Human Tracker results (table 2.44)                     |
| 9 – 15  |               | currently not defined                                  |

Table 2.51: Description of the parameter *autoMeasType*

### 2.8.2 Reception of Measurement Data

The data transmission to the connected TCP sockets in Automatic Operation mode always starts with a synchronization pattern (see table 2.52). This sequence is required for synchronization between radar and host and can be used to access the transmitted measurement data without requesting parameters from the radar module.

| Name         | Data Type | Value                                     |
|--------------|-----------|-------------------------------------------|
| SyncWord1    | Uint16    | 0xAAAA                                    |
| SyncWord2    | Uint16    | 0x5555                                    |
| autoMeasType | Uint16    | Type of transmitted data (see table 2.51) |

Table 2.52: Synchronization pattern of the automatic operation

The synchronization pattern is followed by user defined measurement data specified by the bits of the parameter *autoMeasType*. All types of measurement data (except the header data) are sent exactly like the answer from radar to host in the respective commands. That means that e.g. the time domain data is sent corresponding to the definition in table 2.33. The optional header requested by enabling bit number 4 is defined in table 2.53.

| Name                | Data Type | Description                                                                                           |
|---------------------|-----------|-------------------------------------------------------------------------------------------------------|
| MinFreq             | Uint16    | Start frequency [MHz]                                                                                 |
| ManualBW            | Uint16    | Bandwidth [MHz]                                                                                       |
| TRamp               | Uint16    | Ramp time [ms]                                                                                        |
| Attenuation         | Uint8     | Number representing an attenuation value                                                              |
| RangeBin/DopplerBin | Uint32    | Size of a range bin in [ $\mu\text{m}$ ] for FMCW or size of a doppler bin [ $\mu\text{m/s}$ ] for CW |
| FreqPoints          | Uint16    | Number of frequency domain values                                                                     |
| ZeroPad             | Uint8     | Zero pad factor                                                                                       |
| Norm                | Uint8     | Signal normalization                                                                                  |

Table 2.53: Description of the automatic operation header data

Table 2.54 shows a simplified program flow of the Automatic Operation mode inside the radar module. It becomes clear that a measurement is performed only if at least one socket is connected and the same measurement data is successively sent to every connected socket. The delay time defined by the parameter *autoDelay* is only applied if the measurement and data transfer processes took less time than its value. The more sockets are connected to a host, the longer the data transfer time will be.

```

while true:
    get time1
    if (at least one socket with a fixed port is connected):
        perform measurement defined by bits 0 and 1
        for (every connected socket with fixed port number):
            send sync. pattern: 0xAAAA, 0x5555, autoMeasType
            send data defined by bits 2 - 8
    if (default socket with user defined port is connected):
        perform requested user command
    get time2
    dt = time2 - time1
    if (dt < autoDelay):
        wait for (autoDelay - dt) milliseconds

```

Table 2.54: Simplified program flow of the automatic operation mode

So far so good, but for what kind of application can this Automactic Operation mode be used? Below, the most relevant properties of this mode are summarized:

- All kinds of data generated by the radar module can be transmitted.
- A fixed time can be adjusted between the measurements, provided that the time for a measurement and the data transmission is smaller.
- The time for a measurement and the data transmission is much smaller than with the command execution protocol, because no request or acknowledge messages are sent which are also dependent on the networks latency.
- More than one host can connect to one radar module, but keep in mind that the data transmission needs more time.
- Reading the data stream and working with the data is possible without knowing the radar parameters, because the sync. pattern and the header data provide all necessary values.

A possible scenario could be to perform an FMCW measurement with an UP-Ramp every 100ms and to save the transmitted FD data of all four channels to a file. Another application could be to observe an area using the Human Tracker algorithm. Therefore, a radar measurement and the target tracking process is performed four times a second. Up to six different hosts can connect to the module to get the results but just one host can change some parameters via the default socket.

## 2.9 HTTP Server

This section describes the HTTP server functionality of the sentire<sup>TM</sup> Radar Module which is only available for the Ethernet interface. The radar runs a small HTTP server on the default port 80 and will answer on HTTP requests. Supported are only HTTP versions 1.0 and 1.1.

### 2.9.1 Request Radar Information

If a HTTP request with GET method and the radars IP address as URL is sent to the module, it will answer with a HTML page which can be displayed by any browser. The page shows module specific information as the firmware version, radar number and some other values and also a link button to another page where the Ethernet parameters can be changed.

### 2.9.2 Request HTML page for changing Ethernet parameters

If a HTTP request with GET method and URL [IP Address]/ChangeEthPars is sent or the link button in the information page is used, a HTML page with options to change the Ethernet parameters of the module is sent back.

### 2.9.3 Request change of IP and Port

The current Ethernet parameters IP address, port number, subnet mask and standard gateway of the module can be changed via a HTTP POST request. The URL for this request has to be [IP Address]/SetEthPars and in the message body the radar expects an array of **Uin16** with the values shown in Table 2.55. The words must be sent with LSB first.

| No. | Data Type | Value  | Description                 |
|-----|-----------|--------|-----------------------------|
| 1   | Uint16    | 0xABCD | key value for start of data |
| 2   | Uint16    | 10     | length of data in bytes     |
| 3   | Uint16    | IP1    | part 1 of IP address        |
| 4   | Uint16    | IP2    | part 2 of IP address        |
| 5   | Uint16    | IP3    | part 3 of IP address        |
| 6   | Uint16    | IP4    | part 4 of IP address        |
| 7   | Uint16    | Port   | Port number                 |
| 8   | Uint16    | SM1    | part 1 of subnet mask       |
| 9   | Uint16    | SM2    | part 2 of subnet mask       |
| 10  | Uint16    | SM3    | part 3 of subnet mask       |
| 11  | Uint16    | SM4    | part 4 of subnet mask       |
| 12  | Uint16    | SG1    | part 1 of standard gateway  |
| 13  | Uint16    | SG2    | part 2 of standard gateway  |
| 14  | Uint16    | SG3    | part 3 of standard gateway  |
| 15  | Uint16    | SG4    | part 4 of standard gateway  |

Table 2.55: Payload of SetEthPars POST request (all values must be Uint16 with LSB first).

The radar module will answer with *Content-Type: multipart/report* and a string in the message body. If receiving of values was successful, the answer will be like "Set IP address to ..., port number to ..., subnet mask to ... and standard gateway to ...". If an error occurred, the message will start with "Error...".

### 2.9.4 Request Range Finder Stream

A GET request with URL [IP Address]/StartRangeFinder will start a server sent event (SSE) stream. The radar will enter a self triggered operation mode (2) performing Range Finder measurements as long as the socket connection is established. Below an example for the start of the stream is shown.

**Answer:**

```
HTTP/1.1 200 OK
Content-Type: text/event-stream
Cache-Control: no-cache
Connection: keep-alive
```

```
:
```

```
event: RangeFinderTarget
data: 2034
data: 2388629
data: -4
```

```
event: RangeFinderTarget
data: 2030
data: 16751002
data: -3
```

```
...
```

The start of the stream is indicated by the Content-Type attribute and that the socket should stay connected. The radar only sends events if a range finder target was found. If nothing was found a “:” is sent, which should be treated as comment, to prevent timeouts. If the Range Finder algorithm found a target an event named *RangeFinderTarget* is sent which is followed by 3 data entries. The first entry represents the target range in [mm], the second entry the target angle in [rad] (IQ25 representation) and the last entry the target speed in [mm/s]. Between the measurements, the *autoDelay* of the operation parameters (see table 2.16) is used.

Once the connection is aborted, the radar will change to normal operation mode itself. Also the standard port can be connected with in this mode. The server sent event stream is compatible with the most browsers (except Internet Explorer) but since there will be no HTML page sent, calling this URL would just start the stream. This streaming functionality is very easy to implement and also very fast.

### 2.9.5 Request Human Tracker Stream

A GET request with URL [IP Address]/StartHumanTracker will start a server sent event (SSE) stream. The radar will enter a self triggered operation mode (2) performing Human Tracker measurements as long as the socket connection is established. Below an example for the start of the stream is shown.

**Answer:**

```
HTTP/1.1 200 OK
Content-Type: text/event-stream
Cache-Control: no-cache
Connection: keep-alive
```

```
:
```

```
event: HumanTrackerTargets
data: 2
data: 1,4978,10444862,1133128
data: 2,7685,15208526,339202
```

```
event: HumanTrackerTargets
data: 1
data: 1,9151,-16432065,180023
```

```
...
```

The start of the stream is indicated by the Content-Type attribute and that the socket should stay connected. The radar only sends events if one or more Human Tracker targets/detections (dependent on tracker enabled bit) were found. If nothing was found a “:” is sent, which should be treated as comment, to prevent timeouts.

If the Human Tracker algorithm found a target, an event named *HumanTrackerTargets* is sent which is followed by data entries. The first entry indicates how many targets/detections were found. For each found target/detection another data line will follow. Each line has consists of 4 different numbers separated by “,”. The first number stands for the target ID number, the second number for the target distance in [mm], the third number for the angle in [rad] (IQ25 representation) and the last number for the linear target level. Between the measurements, the *autoDelay* of the operation parameters (see table 2.16) is used.

Once the connection is aborted, the radar will change to normal operation mode itself. Also the standard port can be connected with in this mode. The server sent event stream is compatible with the most browsers (except Internet Explorer) but since there will be no HTML page sent, calling this URL would just start the stream. This streaming functionality is very easy to implement and also very fast.

## 2.10 Software Extension

The functionality built into the sentire™ Radar Module can optionally be extended by user defined software. Within the software extension, the user can:

- Receive parameters from the host,
- Perform measurements,
- Apply the built-in FFT algorithm to the measurement results,
- Apply self-defined data processing,
- Send the results to the host.

The extension software is invoked on the host PC via the command CMD\_EXTENSION (see table 2.56). The communicated data has to match the communication programmed in the extension software. First (if needed) the host sends data to the radar, the radar performs the measurement and processing and returns the result. This section describes how to implement, build and flash the extension software into the radar module.

|                                            |                                  |             |
|--------------------------------------------|----------------------------------|-------------|
| Command Name                               | CMDID_EXTENSION                  |             |
| Command ID                                 | 0x00FF                           |             |
| Description                                | Performs a user defined function |             |
| <i>Data transmitted from Host to Radar</i> |                                  |             |
| Name                                       | Data Type                        | Description |
| Any combination of user defined parameters |                                  |             |
| <i>Data transmitted from Radar to Host</i> |                                  |             |
| Name                                       | Data Type                        | Description |
| Any combination of user defined values     |                                  |             |

Table 2.56: Description of the command CMDID\_EXTENSION

### 2.10.1 Software Requirements

For building extension software, the Code Composer Studio 3 (CCS3) as well as the TI Code Generation Tools with the program “hex2000.exe” are required. Additionally, some source and command files are needed to develop extension functions which are compatible with the standard radar firmware. These files are provided by IMST and are listed in table 2.57.

| File Name          | Description                                                    |
|--------------------|----------------------------------------------------------------|
| Extension.pjt      | Project file (CCS3 format)                                     |
| Extension.cmd      | Linker command file                                            |
| Extension.h        | Header file                                                    |
| ExtensionMain.c    | Main source file                                               |
| ExtensionTable.asm | Software entry point table                                     |
| out2hex.bat        | Invoking tools for generating the hex file                     |
| rts2800_fpu32.lib  | Floating point arithmetic library (optional)                   |
| IQmath_fpu32.lib   | Library for C2000 IQ math floating point operations (optional) |

Table 2.57: List of files required for the development of extension functions

### 2.10.2 Building and Flashing

Before starting the development of user defined functionality, all provided files are copied to a project directory. The project is started by opening the file “Extension.pjt” inside the CCS3 development tool. The user fills his own code into the function body of *ExtensionMain* within the file “ExtensionMain.c” (refer to table 2.57). Additional source files and libraries can also be added to the project. The compilation and build process performed by CCS3 generates a warning “no suitable entry-point found”, because the extension is not stand-alone executable, but has to be called by the standard radar software. This warning can be neglected.

The CCS3 build creates the output file “Extension.out”. CCS3 can upload this file into the radar’s memory, but the user must ensure that only the sector intended for the extension software is flashed, otherwise the standard software will be erased. The upload feature of the Firmware Update GUI erases only the used sectors, so the standard software remains untouched. For flashing by the GUI the output file has to be converted into the Intel HEX format using the tool “hex2000.exe”. The batch file “out2hex.bat” provides all needed parameters and generates a HEX file named “Extension.hex”. If necessary, the local path for “hex2000.exe” has to be adapted inside the “out2hex.bat”.

### 2.10.3 Interface to Standard Software

The function *ExtensionMain* is invoked by the standard software, when the user command CMDID\_EXTENSION (see table 2.56) is received. Hereby, a pointer to the data structure EXTENSION\_PARAMS is passed to the function, providing access to some fundamental functions and parameters of the standard software. The elements of the structure EXTENSION\_PARAMS are defined by table 2.58.

| Element                | Description                                                    |
|------------------------|----------------------------------------------------------------|
| txbyte, txword, txlong | Pointers to transmission functions (8-bit, 16-bit, and 32-bit) |
| rxbyte, rxword, rxlong | Pointers to reception functions (8-bit, 16-bit, and 32-bit)    |
| meas_tddata            | Pointer to the measurement function (time domain data)         |
| calc_fddata            | Pointer to the function calculating the frequency domain data  |
| config                 | Pointer to the configuration data (refer to table 2.59)        |

Table 2.58: Elements of the structure EXTENSION\_PARAMS

The configuration parameters contained in the element *config* of the structure EXTENSION\_CONFIG are listed in table 2.59. Please note, that these parameters are just copies of the corresponding values in the standard software. An alteration has no effect, for the extension they are considered as read only. The values of these parameters can be changed using the standard command CMDID\_SETUP (see section 2.5.10).

| Name        | Data Type | Description                               |
|-------------|-----------|-------------------------------------------|
| fftDataType | UInt8     | Type of transmitted frequency-domain data |
| band        | UInt8     | Bandwidth selection (see table 2.21)      |
| tRamp       | UInt8     | Ramp time in [ms]                         |
| pad         | UInt8     | Zero pad factor (1 = no padding, 2, 4, 8) |
| norm        | UInt8     | Flag: signal normalization enabled        |
| freqPoints  | UInt16    | Number frequency-domain data samples      |
| numActChan  | UInt16    | Number of active IF channels              |
| minFreq     | UInt16    | Lowest ramp frequency in [MHz]            |
| manualBW    | UInt16    | Custom bandwidth in [MHz]                 |
| Attenuation | UInt8     | Attenuation value in [dB]                 |

Table 2.59: List of system parameters made accessible by the structure EXTENSION\_PARAMS (for details refer to table 2.20)

### 2.10.4 Communication Functions

With the communication function pointers listed in table 2.58, the extension software can access the standard functions for communication between radar and host without considering the actual communication interface. They have two input parameters: a data pointer and the desired number of elements. The receive functions return the actual number of received elements. In order to abbreviate code, the following data types are introduced:

- unsigned char  $\rightarrow$  u08
- unsigned int  $\rightarrow$  u16

- unsigned long  $\rightarrow$  u32

The declaration of the data transfer functions is listed in table 2.60.

| Declaration                          | Description                                  |
|--------------------------------------|----------------------------------------------|
| void (*txbyte)(const u08 *p, u16 n); | send byte array of $n$ elements              |
| u16 (*rxbyte)(u08 *p, u16 n);        | receive byte array of $n$ elements           |
| void (*txword)(const u16 *p, u16 n); | send 16-bit integer array of $n$ elements    |
| u16 (*rxword)(u16 *p, u16 n);        | receive 16-bit integer array of $n$ elements |
| void (*txlong)(const u32 *p, u16 n); | send 32-bit integer array of $n$ elements    |
| u16 (*rxlong)(u32 *p, u16 n);        | receive 32-bit integer array of $n$ elements |

Table 2.60: Declaration of data transfer functions

### Examples

Sending a byte array to host ( $p$  is the name of the pointer to the parameter structure):

```
u08 b_arr[7];
u16 n_send = 7;
... // fill array with data
p->txbyte(b_arr, n_send);
```

Casting an array of *signed* characters:

```
char b_arr[7];
u16 n_send = 7;
... // fill array with data
p->txbyte((u08*)b_arr, n_send);
```

Accessing a single byte using the address operator (&):

```
u08 b;
u16 n_send = 1;
...
p->txbyte(&b, n_send);
```

Using the receive function and getting the actual number of received elements:

```
u08 b_arr[7];
u16 n_rec_desired = 7;
u16 n_rec_actual;
n_rec_actual = p->rxbyte(b_arr, n_rec_desired);
```

The functions for word (16-bit) and long (32-bit) arrays are invoked in the same way. The communication may be in one direction (reception or transmission only) or both (reception

and transmission). As stated above, the communication protocol is designated for first sending data from host to radar and then opposite. So once a transmission function is called, no further call of a reception function is possible.

### 2.10.5 Measurement and Data Processing

The measurement is performed by the function *meas\_tddata* (see table 2.58). The passed parameter denotes the ramp direction and is defined in table 2.61.

| Ramp Code     | Numerical Value | Description               |
|---------------|-----------------|---------------------------|
| RAMP_DIR_GAP  | 0               | CW measurement            |
| RAMP_DIR_UP   | 1               | FMCW, Up frequency ramp   |
| RAMP_DIR_DOWN | 2               | FMCW, Down frequency ramp |

Table 2.61: Possible states of the ramp direction (passed to function *meas\_tddata*)

The return value is a two-dimensional 32-bit array with the measurement results. The first dimension is the channel number (the total number of channels is given in the configuration as *numActChan*). The second dimension is the measurement index (1024 values). The measurement values are given as signed long numbers. The result of *meas\_tddata* represents the Time Domain data (TDD). The measurement parameters can be obtained from the configuration parameters *tRamp*, *minFreq* and *manualBW* (table 2.59).

The standard software provides an FFT with various data representation styles. The configuration parameter *fftDatType* defines the representation style (refer to table 2.62).

| Data Type     | Numerical Value | Description                      |
|---------------|-----------------|----------------------------------|
| DT_MAGN       | 0               | Magnitude only                   |
| DT_MAGN_PHASE | 1               | Magnitude and phase [rad]        |
| DT_REAL_IMAG  | 2               | Real and imaginary part          |
| DT_MAGN_ANGLE | 3               | Magnitude and object angle [rad] |

Table 2.62: Possible states of the parameter *fftDatType*

To perform the FFT, the function *calc\_fddata* has to be invoked (see table 2.58). The function processes the data of the last measurement, so it may not be called before the first measurement. The result is the Frequency Domain data (FDD) in the form as defined by *fftDatType* given as a two-dimensional array of complex 32-bit numbers. The number of valid FD data points is defined by the value of *freqPoints* (see table 2.59). Since FDD and TDD use the same memory space, the TDD is no more available after FFT.

**Example:**

```
long **tdd;
CMPLX32 **fdd;
// Perform measurement with Up ramp
tdd = p->meas_tddata(RAMP_DIR_UP);
// tdd[i][j] represents measurement value j of channel i
// Perform FFT
fdd = p->calc_fddata();
// fdd[i][j] represents FDD value j of channel i
// real and imaginary part are fdd[i][j].real and fdd[i][j].imag
// tdd is not longer valid
```

# Appendix

## A Command Details

### A.1 Set/Get System Parameters

The values of the parameters *Band*, *TRamp* and *ZeroPad* (see table 2.20) are affecting the resolution in distance and the resolution in speed of the frequency-domain data. The difference of the corresponding distances between two consecutive FD values of a FMCW measurement is further named *Range bin*. The difference of the corresponding velocities between two consecutive FD values of a CW measurement is further named *Doppler bin*.

The size of a Range bin can be calculated by

$$\delta_r = \frac{c_0}{2 \cdot B} \quad (3.1)$$

|              |                                                               |
|--------------|---------------------------------------------------------------|
| $\delta_r$ : | size of a Range bin                                           |
| $c_0$ :      | speed of light                                                |
| $B$ :        | bandwidth of the frequency ramp (parameter <i>ManualBW</i> ). |

The Radar Module recalculates the value of a Range bin (parameter *RangeBin*) after setting new system parameters and returns the new value with command 0x0029. It should be mentioned that the value returned by the Radar Module differs slightly from the theoretical value because some hardware specific timings are taken into account for the calculation. The corresponding distance to a FD value can be calculated with:

$$d[n] = (n - 1) \cdot \frac{RangeBin}{ZeroPad \cdot 10^6} \quad (3.2)$$

|                 |                                                       |
|-----------------|-------------------------------------------------------|
| $d[n]$ :        | distance [m] value of the nth FD value                |
| $n$ :           | number of the FD value ( $1 \leq n \leq FreqPoints$ ) |
| <i>RangeBin</i> | parameter <i>RangeBin</i> [ $\mu$ m]                  |
| <i>ZeroPad</i>  | parameter <i>ZeroPad</i> .                            |

The size of a Doppler bin can be calculated by

$$\delta_v = \frac{c_0}{2 \cdot f_0 \cdot T} \quad (3.3)$$

|              |                                              |
|--------------|----------------------------------------------|
| $\delta_v$ : | size of a Doppler bin                        |
| $c_0$ :      | speed of light                               |
| $f_0$ :      | signal frequency (parameter <i>MinFreq</i> ) |
| $T$ :        | sample time.                                 |

The original sample time differs slightly from the time specified by the parameter *TRamp* because of the frequency of the used Analog-to-Digital Converter. The real time the FMCW or CW signal is sampled is  $0.98304 \text{ ms} \cdot t_R$  where  $t_R$  is the value of the parameter *TRamp*.

As for the parameter *RangeBin*, the parameter *DopplerBin* is recalculated after setting new system parameters. The corresponding velocity to a FD value can be calculated with:

$$v[n] = (n - 1) \cdot \frac{\text{DopplerBin}}{\text{ZeroPad} \cdot 10^6} \quad (3.4)$$

|                     |                                                              |
|---------------------|--------------------------------------------------------------|
| $n$ :               | number of the FD value ( $1 \leq n \leq \text{FreqPoints}$ ) |
| $v[n]$ :            | velocity [m/s] value of the nth FD value                     |
| <i>DopplerBin</i> : | parameter <i>DopplerBin</i> [ $\mu\text{m/s}$ ]              |
| <i>ZeroPad</i> :    | parameter <i>ZeroPad</i> .                                   |

## A.2 Get Time-Domain Data after UP-Ramp

On command 0x0045 (see chapter 2.5.22), the Radar Module performs an UP-Ramp and samples the time-domain-signal. The sampled data for each activated channel is transmitted to the control-system in a data-block corresponding to table 2.33.

To calculate a voltage level from a time-domain sample use the following equation:

$$U_{TD}[n] = 3 \text{ V} \cdot \frac{TD[n]}{2^{12} \cdot AVG \cdot t_R}. \quad (3.5)$$

|               |                                                                                      |
|---------------|--------------------------------------------------------------------------------------|
| $n$ :         | index of a TD sample                                                                 |
| $U_{TD}[n]$ : | voltage level of the nth TD sample                                                   |
| $TD[n]$ :     | $n^{\text{th}}$ TD sample                                                            |
| $AVG$ :       | fixed average factor from oversampling (=4)                                          |
| $t_R$ :       | variable average factor which corresponds to the value of the ramp time <i>TRamp</i> |

### A.3 Get Frequency-Domain Data after UP-Ramp

On command 0x0046 (see chapter 2.5.23) the Radar Module performs an UP-ramp and samples the Time-Domain signal. The Radar Module calculates Frequency-Domain data from the sampled time-domain-data and transmits it with additional information to the control-system. The data block is described in table 2.34.

In order to calculate a relative level in [dBm] for each FD magnitude transmitted by the Radar Module, the equation below can be used:

$$LVL[n] = 20 \log\left(\frac{FD[n]}{2^{21}}\right) \quad (3.6)$$

$n$ : index of a FD magnitude value

$LVL[n]$ : relative level of the nth frequency-domain sample in [dBm]

$FD[n]$ :  $n^{\text{th}}$  magnitude value of the FD data.

As described in table 2.34, the type of frequency-domain data received depends on the value of the parameter *FftDatTyp*. If it is set to 2, complex data is transmitted. If it is 1 or 3, the phase respective object angle is transmitted in [rad] and IQ25 representation. These angles can be converted to degrees by

$$\varphi[n] = \frac{\phi[n] \cdot 180^\circ}{2^{25} \cdot \pi} \quad (3.7)$$

$n$ : index of a phase or object angle value

$\varphi[n]$ :  $n^{\text{th}}$  phase or object angle in [deg]

$\phi[n]$ :  $n^{\text{th}}$  phase or object angle value in [rad] (IQ25).

## B Referenced Documents

This following list contains a table of documents referenced by this manual.

| Ref. ID | Document Name                | Source    |
|---------|------------------------------|-----------|
| /RAD/   | sR1200_Radar_Manual.pdf      | IMST GmbH |
| /CMD/   | sR1200_Command_Interface.pdf | IMST GmbH |
| /GUI/   | sR1200_Development_Kit.pdf   | IMST GmbH |

## C List of Abbreviations

|         |                                                     |
|---------|-----------------------------------------------------|
| ADC:    | Analog-to-Digital Converter                         |
| CAN:    | Controller Area Network                             |
| CW:     | Continuous Wave                                     |
| EEPROM: | Electrically Erasable Programmable Read-Only Memory |
| FD:     | Frequency Domain                                    |
| FFT:    | Fast Fourier Transform                              |
| FMCW:   | Frequency Modulated Continuous Wave                 |
| GUI:    | Graphical User Interface                            |
| HT:     | Human Tracker (algorithm)                           |
| ID:     | Identifier                                          |
| IF:     | Intermediate Frequency                              |
| IP:     | Internet Protocol                                   |
| ISM:    | Industrial, Scientific and Medical (frequency band) |
| RF:     | Radio Frequency or Range Finder (algorithm)         |
| Rx:     | Receiver (antenna, etc.)                            |
| SPI:    | Serial Peripheral Interface                         |
| TCP:    | Transmission Control Protocol                       |
| TD:     | Time Domain                                         |
| Tx:     | Transmitter (antenna, etc.)                         |
| USB:    | Universal Serial Bus                                |

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# Important Notices

## Regulatory Compliance Information

The use of radio frequencies is limited by national regulations. The Radar Module has been designed to comply with the European Union's R&TTE (Radio & Telecommunications Terminal Equipment) directive 1999/5/EC and can be used license free within the European Union at 24–24.25 GHz. Nevertheless, please refer to your national frequency regulation if there is any need for a notification at least.

The Radar Module (with housing) has been designed to be used with suitable IT-equipment to collect, store and process the radar data (referred as “final system”). According to the R&TTE directive, the declaration of compliance with the essential requirements of the R&TTE directive is within the responsibility of the manufacturer of the final system. A declaration of conformity (DoC) for the Radar Module is available from IMST GmbH on request.

The applicable regulation requirements are subject to change. IMST GmbH does not take any responsibility for the correctness and accuracy of the aforementioned information. National laws and regulations, as well as their interpretation can vary with the country. In case of uncertainty, it is recommended to contact either IMST's accredited Test Center or to consult the local regulatory authority of the respective country of use.

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